Pool Ball Detection and Recognition

Computer Vision (CSU44053)

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1 Assignment Overview

The objective of this assignment was to develop a computer vision system capable of locating and recognising pool balls from photographs of a pool table. The two main tasks were to determine the position of each ball and to classify it as solid, striped (along with each ball's respective number) or white cue ball.

The dataset consisted of one reference image, AllBalls.jpeg, and six comparison images, Table1–Table6.jpeg, containing different ball arrangements. All images were captured from the same camera viewpoint, and the four corners of the playing surface were provided. AllBalls was used to tune the original detector; the six comparison images were subsequently also used for recognition tuning. The later selected recognition scores are therefore exploratory tuning results.

The system was implemented in C++ using OpenCV, following a classical image-processing approach rather than a neural network. The implemented pipeline combines circular Hough detection, geometric filtering, perspective transformation, colour segmentation and rule-based classification.

The experiments favoured retaining the original detector and using a tuned six-cluster recognition configuration with a targeted solid-ball CCA correction. This combination correctly classified 82/86 matched types and 75/86 numbers. It remains implemented in the experiment executable; the main application still runs the original two-cluster baseline. The report distinguishes these implementation states and explains why padding, cloth removal and alternative segmentation-based detectors were not selected.

2 System Design and Implementation

2.1 Pipeline Overview

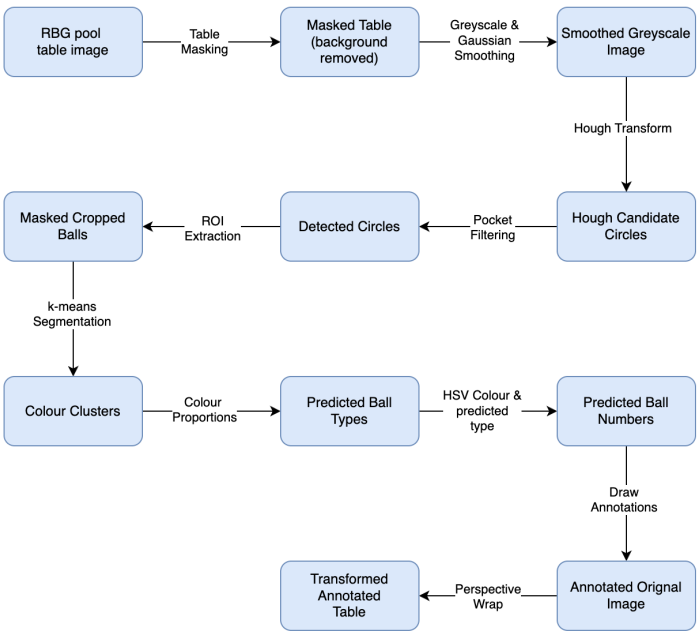


Figure 1: Overall processing pipeline, from input image to detected and classified balls.

The processing pipeline consists of two main stages: ball localisation and ball recognition. Localisation uses geometric and edge information to identify candidate circles. Ball recognition analyses the colour composition of each detected ball. The implementation description below first documents the original baseline; the experimental changes and final recommendation are evaluated separately.

2.2 Ball Detection

Since the table boundaries were known and the camera position remained fixed, the playing surface could be represented by a quadrilateral. A binary mask was constructed from its four corners, restricting circle detection to the relevant image region and reducing interference from the surrounding environment. The goal of this step was to remove the background surrounding the pool table so that it would not interfere with ball detection.

The masked image was converted to grayscale and smoothed using a 5×5 Gaussian filter with a standard deviation of 1.5. Smoothing was used as it reduces high-frequency noise and small local intensity variations that might otherwise introduce spurious edge responses. These preprocessing settings were fixed during the Hough parameter search; the experiment does not establish that they are independently optimal.

Candidate ball locations were detected using OpenCV's circular Hough transform with the gradient-based method (`HOUGH_GRADIENT`). Applied to the grayscale, Gaussian-smoothed image, it identifies circular structures and returns their estimated centres and radii. These circles were treated as candidates because other features, such as pockets, could also produce detections.

2.2.1 Parameter tuning using grid search

The Hough transform requires several parameters that affect how many circles are detected and how accurately they match the balls. Rather than choosing these values only by visual inspection, a grid search was used to compare parameter combinations on the masked tuning image, `AllBalls.jpeg`.

The search varied four parameters:

- Accumulator resolution ratio: $dp \in \{1.0, 1.5, 2.0\}$.
- Minimum distance between detected centres: $\{15, 18, 20, 22, 25\}$ pixels.

- Upper Canny edge threshold: $\{75, 100, 125, 150\}$.
- Circle accumulator threshold: $\{15, 20, 25, 30, 35, 40, 45, 50\}$.

This produced 480 parameter combinations. The minimum and maximum circle radii were fixed at 15 and 30 pixels respectively.

For each combination, detected centres were transformed into table coordinates and matched to the annotated centres within a tolerance of 15 pixels. The best configuration was selected by first maximising the number of true positives, then minimising false positives, and finally minimising the mean centre localisation error.

The results for every combination were saved to a CSV file, together with an image showing the detections from the best configuration. Pocket filtering was not applied during this search; it was evaluated as a separate processing step.

2.2.2 Removing pocket detections

Table pockets can also produce circular detections. To remove these, candidate centres were transformed into table coordinates and compared with six approximate pocket positions. Candidates within 25 pixels of a pocket centre were discarded.

After choosing the Hough parameters and applying pocket filtering, we obtained the following results on the tuning image:



Figure 2: AllBalls Ball detection with final Hough Params.

2.3 Ball Extraction and Colour Segmentation

Once the candidate balls have been detected, each ball is extracted from the original colour image using a square region centred on the predicted circle. A circular binary mask restricts subsequent processing to pixels belonging to the ball.

Initial experiments investigated colour thresholding, including HLS-based methods. However, a single global threshold was difficult to apply consistently across differently coloured balls, particularly in the presence of highlights, shadows and white markings.

The original baseline implementation uses K-means clustering to partition the colour pixels of each ball into two groups. Clustering is performed on BGR pixel values, and the resulting cluster centres are converted to HSV for interpretation.

The later recognition search evaluated K=2–6. Its recommended K=6 setting uses the same BGR clustering method but a more detailed colour partition and different interpretation thresholds. A colour cluster is not necessarily a spatially connected region; connected regions are measured separately by CCA.

The proportion of ball pixels assigned to each cluster is calculated. Each cluster's representative colour is then converted to HSV and categorised using the following rules:

- **Light:** brightness $V \geq 170$ and saturation $S \leq 165$.
- **Dark:** brightness $V \leq 90$.
- **Coloured:** all remaining clusters.

Here, saturation and brightness range from 0 to 255. The percentages of clusters in each category are added together to obtain the total light, dark and coloured proportions used for ball-type classification.

This approach reduces the original pixel-level information to a small set of interpretable features. It does not require manually defining a separate colour threshold for every individual ball.

2.4 Ball recognition

Ball recognition is performed in two stages. First, each detected ball is classified as white, solid or striped. Its predicted type is then combined with its colour to estimate its number.

2.4.1 Ball Type classification

Classification is performed using the proportions of light, dark and coloured pixels obtained from segmentation.

A ball is classified as white when at least 85% of its pixels belong to light clusters. It is classified as solid when at least 70% of its pixels belong to non-light clusters. Balls containing both light and non-light regions, without meeting the preceding criteria, are classified as striped.

These rules reflect the visual characteristics of pool balls: solid balls are predominantly coloured, while striped balls contain a substantial white region in addition to their coloured band.

However, this approach is sensitive to the visible orientation of the ball. A striped ball viewed from its white side may resemble the cue ball, while one viewed predominantly from its coloured side may resemble a solid ball.

An additional experimental classifier was developed using connected-component analysis. It examines the spatial organisation of light and non-light regions, including the largest connected components and the proportion of the circular boundary occupied by light pixels. Its performance is compared with the simpler colour-based method in Section 3.

2.4.2 Ball number recognition

Number recognition uses the predicted ball type and the HSV colour of the largest eligible non-light cluster. Clusters covering less than 2% of the ball area are ignored, as are light clusters ($V \geq 170$ and $S \leq 165$).

A ball predicted as white is assigned number 0, representing the cue ball. For solid and striped balls, the selected cluster is examined using the following rules. OpenCV represents hue on a scale from 0 to 179, while saturation and brightness range from 0 to 255.

If $S < 60$, the ball is assigned number 8 only when $V \leq 90$ and its predicted type is solid. Otherwise, this low-saturation case returns an unknown result.

For clusters with $S \geq 60$, the colour thresholds in Table 1 determine the base number.

Table 1: Original baseline HSV thresholds for ball-number recognition.

Colour	Base number	HSV condition
Yellow	1	$15 \leq H < 35$
Blue	2	$100 \leq H < 130$
Red	3	($H < 6$ or $H \geq 170$), $V \geq 150$
Purple	4	$130 \leq H < 165$
Orange	5	$6 \leq H < 15$
Green	6	$45 \leq H < 90$
Maroon	7	($H < 6$ or $H \geq 170$), $V < 150$

Solid balls retain the base number, while striped balls add 8, giving numbers 9–15. Red and maroon share the same hue range, so brightness is used to distinguish them.

When segmentation is unavailable, the predicted type is unknown, or no eligible cluster or colour match is found, the predicted number is set to -1. These thresholds are experimental and may be sensitive to changes in lighting.

2.5 Perspective Transformation

The table appears at an angle in the original photographs, while the provided ground-truth positions refer to a rectangular, top-down view. To compare our detections with these positions, their coordinates must be converted into the same coordinate system.

The four known table corners were mapped to the corners of a 1434×683 pixel rectangle. The resulting perspective transformation was then applied to each detected ball centre.

Ball detection and recognition were performed on the original image. Transforming the image beforehand could stretch the circular ball outlines into ellipses, making the Hough circle detector less reliable. The full image was transformed separately when producing visualisations for comparison with the ground truth.

3 Evaluation and Results

3.1 Methodology

For the original detector and baseline evaluation, the reference image AllBalls.jpeg was used for parameter selection, while the six other images were reserved for evaluation. The later exploratory recognition search in Section 3.4 also uses these six images for tuning; its results are reported separately. The original separation reduces the risk of reporting performance only on an image used to optimise the detector.

An automated grid search evaluated 480 combinations of Hough parameters. Configurations were ranked by the number of correctly detected balls, followed by the number of false positives and the average localisation error.

The final parameters were:

Table 2: Final Hough transform parameters for ball detection.

Parameter	Value	Description
`dp`	1.5	Image-to-accumulator resolution ratio
`minDist`	18	Minimum distance between centres (pixels)
`param1`	75	Upper Canny edge threshold
`param2`	35	Accumulator threshold for accepting circles
Minimum radius	15	Smallest accepted circle radius (pixels)
Maximum radius	30	Largest accepted circle radius (pixels)

Detected centres were compared with the provided ground truth after perspective transformation. A one-to-one matching procedure assigned detections to annotated centres within a maximum distance of 15 pixels in table coordinates.

Unmatched detections were counted as false positives (FP), while unmatched ground-truth balls were counted as false negatives (FN). Correctly matched detections were counted as true positives (TP).

Performance was assessed using precision, recall and F1-score. The mean Euclidean distance between matched centres was also calculated to evaluate localisation accuracy.

3.2 Ball Detection Results

In the original tuning run, the selected configuration detected all 16 annotated balls, with three additional false detections. This gave a precision of 84.21%, recall of 100% and F1-score of 91.43%, before pocket filtering.

For the final experiment comparison, the current full masked and pocket-filtered detector was rerun consistently on all six comparison images. These current measurements replace the older saved validation snapshot in the headline results. That historical CSV recorded seven false detections, whereas the consistent reruns used by the later experiments recorded two. The report does not attribute this difference to a specific change without evidence.

Table 3: Current standard-detector baseline rerun on Table1–Table6. Matching uses a 15-pixel tolerance in table coordinates.

Image	GT	TP	FP	FN	F1	Mean error
Table 1	16	14	1	2	0.9032	7.68
Table 2	16	16	1	0	0.9697	3.58
Table 3	16	15	0	1	0.9677	3.12
Table 4	15	14	0	1	0.9655	3.89
Table 5	14	14	0	0	1.0000	5.83
Table 6	13	13	0	0	1.0000	5.76
Total	90	86	2	4	0.9663	4.91

Across all six images, the rerun detector achieved precision 97.73%, recall 95.56% and F1-score 96.63%, with 86 true positives, two false positives and four misses among 90 annotations. The mean localisation error across matched detections was approximately 4.91 table pixels. These are the baseline counts used for all subsequent segmentation-detector comparisons.

The relatively high recall indicates that the method successfully identifies most of the visible balls. However, its performance is less consistent when balls overlap or are closely grouped.

Table 1 produced the lowest detection F1-score and the largest mean centre error (7.68 pixels). This image contains a tightly arranged group of balls, where adjacent circular boundaries may interfere with Hough detection and accurate localisation.

By contrast, the later images contain more spatially separated balls, making individual circular outlines easier to identify. This is consistent with the generally stronger detection results on Tables 3–6.

3.3 Ball Recognition Results

Recognition was evaluated using only detections successfully matched to ground-truth balls. This conditions classification on a successful detection match and avoids counting missing balls as classification errors. It does not eliminate the effect of inaccurate centres or contaminated crops. The following table describes the original baseline and untuned CCA; the later experiments improve on these recognition settings.

The baseline K-means colour classifier correctly classified 76 of the 86 matched balls, giving an accuracy of 88.37%.

The experimental classifier incorporating connected-component features correctly classified 73 of the 86 matched balls, reducing accuracy to 84.88%.

Table 4: Original recognition controls on the 86 matched balls in Table1–Table6.

Method	Correct type	Type accuracy	Correct number	Number accuracy
K=2 colour rules	76/86	88.37%	67/86	77.91%
K=2 + original CCA	73/86	84.88%	64/86	74.42%

The additional connected-component rules introduced three new classification errors without correcting any existing errors on these comparison images.

This suggests that spatial information is not necessarily more reliable than overall colour proportions. In particular, the shape and boundary coverage of white regions vary considerably with the orientation of striped balls. Consequently, rules intended to recognise stripes can also incorrectly modify predictions for the white cue ball or solid balls.

The simpler colour-based method remained in the main application. This result does not rule out all CCA corrections: the later targeted solid-only rule is evaluated separately in Section 3.5.

The recognition accuracy of 88.37% is conditional on successful detection. Considering all 90 ground-truth balls, the system correctly detected and classified 76, corresponding to an end-to-end correct identification rate of 84.44%.

This distinction is important because a classification system cannot correctly identify a ball that was missed during localisation.

3.4 Exploratory Recognition Experiments

The baseline results above describe the original implementation. Two subsequent experiments investigated whether the extraction mask and recognition parameters could improve these results. Both experiments were implemented in a separate `experiments/` directory and executed using the independent `BallExperiments` target. They did not replace the classification functions, configuration values, or execution path of the main application. The results in this subsection therefore describe experimental alternatives rather than a change to the deployed baseline.

3.4.1 Evaluation protocol and interpretation

For these experiments, Table1–Table6 were used as an exploratory tuning set. AllBalls was reported separately as a reference image and was not included in the ranking objective of the recognition grid search. This differs from the earlier detection experiment, which selected Hough parameters on AllBalls and evaluated them on Table1–Table6. Since the recognition search used the labels of Table1–Table6 to select thresholds, its final scores on those images are tuning scores, not independent test estimates. Furthermore, all seven images share the same viewpoint and some unchanged ball positions. AllBalls is also the image on which the original rules were developed. Its separate results are consequently a useful regression check, but not evidence of generalisation to a new camera, table or lighting condition.

The detector, pocket filtering, perspective transformation and matching tolerance were held fixed. Detection was performed once per image. Predicted centres were transformed into table coordinates and matched one-to-one to annotated ball centres with a maximum distance of 15 table pixels, using the existing evaluation implementation. A match was established before varying recognition parameters and reused for every configuration. The experimental run contained 90 annotated balls in Table1–Table6, of which 86 were matched. There were two unmatched candidate detections, which were excluded from recognition accuracy and displayed with an unavailable correctness indicator. All 16 annotated balls in AllBalls were matched. These are counts measured by the experiment's actual detection run; they are not inferred from a previously saved detection table.

For a matched ball, its annotated number determines its true type: number 0 denotes the white cue ball, numbers 1–8 are solid, and numbers 9–15 are striped. Candidate indices identify detections within an image and should not be interpreted as ball numbers.

Four measures were recorded. Let M be the number of matched detections, G the number of annotated balls, t_i the true type and n_i the true number of matched candidate i. Let $\hat{t}_i$ and $\hat{n}_i$ be the predicted type and number. The recognition accuracies are

$$A_{\text{type}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}(\hat{t}_i = t_i), \qquad A_{\text{number}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}(\hat{n}_i = n_i).$$

The number prediction uses the predicted type, so a type error can propagate into number recognition. An additional diagnostic reruns number classification with the true type supplied:

$$A_{\text{oracle}} = \frac{1}{M}\sum_{i=1}^{M}\mathbf{1}\big(f_{\text{number}}(\text{clusters}_i, t_i) = n_i\big).$$

This diagnostic helps separate colour-selection failures from type errors. It is not an executable recognition solution because ground truth is required. In particular, supplying the true type of a cue ball invokes the

existing number-0 shortcut when segmentation is available. Thus the measure should not be described as a pure hue-classification accuracy. Finally, end-to-end number recall divides correct detected-and-numbered balls by G instead of M. Missed balls remain errors under this measure. It does not measure the precision of numbered detections, since false-positive candidates are not included in its numerator or denominator.

3.4.2 Radius padding and cloth removal

The initial hypothesis was that Hough detection could underestimate a ball's radius and exclude useful pixels around its edge. The extraction radius was therefore changed from the rounded Hough radius r to $r' = r + p$, where $p \in \{0, 1, 2, 3, 4\}$ pixels. Both the square crop and its circular mask used r'. Enlarging the square alone would not admit additional pixels into colour analysis, because pixels outside the original mask would still be excluded. Crops were clipped to the image boundaries. Padding zero without filtering reused the original extraction function and reproduced the baseline mask.

Each padding value was evaluated with and without a local cloth-removal filter, producing ten variants per candidate. K was fixed at 2 and all type and number thresholds retained their original values. This isolated the effect of extraction from changes to the recognition rules.

The filter estimated the surrounding cloth colour from an annulus of width six pixels immediately outside the padded circle. Sampling used the original colour image. A sample was eligible when its OpenCV HSV values satisfied $35 \leq H \leq 95$, $S \geq 50$ and $V \geq 30$. These conditions restricted the estimate to visible green material rather than including arbitrary colours from nearby objects. If fewer than 20 eligible samples were available, filtering was skipped and the padded circular mask was retained.

The modal hue bin of eligible samples was selected. Saturation and brightness were averaged over samples within ten hue units of that mode. Pixels inside the candidate's mask were marked as potential cloth when their hue differed from the modal hue by at most 10, their saturation and brightness differed from the corresponding averages by at most 70, and they satisfied the same minimum saturation and brightness conditions. These tolerances use OpenCV's 0–179 hue and 0–255 saturation/brightness scales.

Potential cloth pixels were divided into eight-connected components. Only components touching the boundary of the circular mask were removed. The boundary was computed by subtracting a 3×3 elliptical erosion from the original mask, with pixels outside the crop treated as zero. This also recognised image-clipped edges as boundaries. Removed pixels were excluded from the mask rather than replaced by black pixels inside the analysis area. Consequently, K-means percentages used only the retained pixels as their denominator. The black background in diagnostic images was for display.

Boundary connectivity reduces the chance of deleting isolated green markings inside another ball, but does not distinguish a green ball from the cloth when their colours are similar and the matching region reaches the boundary. The observed masks showed that real green ball regions could be removed. Conversely, shadows and reflections could prevent some cloth pixels from matching the estimated colour. Cloth removal was therefore treated as an experimental option rather than assumed to be beneficial.

Table 5: Radius-padding experiment on matched balls in Table1–6 using the original recognition thresholds and K=2.

Padding (px)	Cloth filter	Matched	Correct type	Correct number
0	Off	86	76	67
1	Off	86	73	60
2	Off	86	67	49
3	Off	86	62	43
4	Off	86	60	42
0	On	86	77	69
1	On	86	76	66
2	On	86	78	64
3	On	86	72	55
4	On	86	66	50

Without filtering, increasing padding reduced correct number predictions on Table1–Table6 from 67 at zero padding to 60, 49, 43 and 42 at one through four pixels respectively. Cloth filtering partially compensated for the additional background, but padded variants still failed to improve number recognition over the

original baseline. Two-pixel padding with filtering increased correct type predictions from 76 to 78, but reduced correct numbers from 67 to 64. The zero-padding filter alone gave 77 correct types and 69 correct numbers. On AllBalls, the unfiltered baseline already classified all 16 balls correctly; the filter at zero padding reduced both counts to 15.

These findings support retaining the original extraction radius for the next experiment. A likely explanation for the deterioration with padding is that additional cloth pixels alter cluster colours and proportions. This is an interpretation of the masks and recognition changes, not a direct measurement of the geometric accuracy of the Hough radius. The results do not establish that every detected radius equals the true visible ball boundary.

3.4.3 Recognition search: fixed processing and caching

The next experiment varied K, the interpretation of cluster colours and the type and number decision thresholds. Radius padding was fixed at zero. Cloth filtering remained an on/off factor, using the same filter parameters as the preceding experiment. Connected-component type correction was not included, so changes could be attributed to clustering and colour rules.

K-means was performed in three-dimensional BGR space using the existing implementation. The requested cluster counts were $K \in \{2, 3, 4, 5, 6\}$. For each call, the random-number generator state was set to 12345, three K-means++ initialisations were attempted, and the termination criteria were 50 iterations or an epsilon of 0.2. The prior random state was restored afterwards. These controls make comparisons repeatable within the same software environment; bit-for-bit equality across different OpenCV versions or platforms is not assumed.

Each candidate was segmented once for each K/filter combination. The run contained 104 candidates across all images, giving 1,040 cached segmentations. Only cluster percentages and representative BGR/HSV colours were retained for threshold evaluation. Cluster masks and segmented images were released after caching because subsequent rules depend on these features rather than on individual pixel labels. The original masked crops were retained for final diagnostic figures. No K-means rerun was required for each threshold configuration.

The experimental classifiers accepted explicit parameter structures rather than changing constants in the main classifier. Before searching, the experimental default type, percentages and number predictions were compared with the original production functions on every cached segmentation. Any disagreement terminated the experiment. This checked that baseline scores and parameterised scores represented the same original decision rules.

3.4.4 Stage one: joint K and type-threshold grid

The first stage evaluated the full Cartesian product below. The original number thresholds were fixed during this stage, although changing the light thresholds also affected which cluster was eligible for number recognition. The same light definition was used in type classification and body-colour selection.

Table 6: Stage-one recognition search space.

Parameter	Evaluated values
Number of clusters K	2, 3, 4, 5, 6
Cloth filtering	Off, on
Minimum light-cluster V	150, 170, 190
Maximum light-cluster S	125, 145, 165, 185
Minimum light proportion for cue ball	80%, 85%, 90%
Minimum non-light proportion for solid ball	60%, 65%, 70%, 75%, 80%

This produced $5 \times 2 \times 3 \times 4 \times 3 \times 5 = 1{,}800$ configurations. A cluster was light when V met the minimum and S did not exceed the maximum. A ball was first considered for the cue-ball rule, then the solid rule, and finally the striped rule. Percentages and ordering therefore followed the original implementation. A failed segmentation or absence of sufficient evidence remained unknown and counted as incorrect for a matched ball.

The dark-cluster threshold was fixed at $V \leq 90$. Although it changes the separate reported dark and coloured percentages, it does not change a type decision: the solid and striped rules use the sum of these two categories. Searching it as a type parameter would duplicate equivalent predictions. The separate black-ball thresholds in number recognition were searched in stage two.

Configurations were ranked first by correct full-number predictions, then by correct type predictions, then by correct oracle-number predictions. All counts used Table1–Table6 only. Remaining ties favoured the smaller departure from the original parameters. The departure summed absolute threshold changes normalised by their search step sizes, plus $K - 2$ and one unit for enabling cloth filtering. The hue boundary at 165 used a five-unit normalisation even though its upper alternative was 168; the body-area threshold used absolute percentage-point change. Final ties favoured the earlier configuration ID. This tie rule prefers less change when measured performance is equal; it does not establish that the chosen thresholds are uniquely optimal. A separate best-type ranking reversed the first two accuracy criteria.

Table 7: Effect of K and cloth filtering on Table1–6. Default columns use the original thresholds. Tuned columns select the highest-ranked stage-one configuration within each K/filter pair; colour thresholds remain fixed.

K	Cloth	Default type	Default no.	Tuned type	Tuned no.	ID
2	Off	76/86	67/86	80/86	71/86	70
2	On	77/86	69/86	79/86	71/86	249
3	Off	73/86	61/86	81/86	69/86	370
3	On	77/86	67/86	80/86	70/86	610
4	Off	73/86	65/86	79/86	71/86	820
4	On	76/86	67/86	79/86	70/86	1060
5	Off	73/86	65/86	79/86	71/86	1179
5	On	76/86	68/86	79/86	71/86	1359
6	Off	74/86	66/86	81/86	73/86	1510
6	On	76/86	67/86	81/86	72/86	1750

The default-threshold columns show that increasing K alone did not improve number recognition. Without cloth filtering, the original K=2 configuration correctly numbered 67 balls, compared with 61, 65, 65 and 66 for K=3–6. However, after joint type-threshold tuning, K=6 without filtering achieved 73 correct numbers and 81 correct types. Thus the effect of K depended on how the resulting clusters were interpreted; the experiment did not support simply increasing K while leaving all thresholds fixed.

3.4.5 Stage two: grouped colour-threshold search

To limit the much larger colour search, a shortlist was formed from the top three distinct K/filter pairs under the full-number ranking and the top three distinct K/filter pairs under the type-first ranking. The original K=2 baseline, with and without cloth filtering, was also included. Exact duplicate seed configurations were removed. This run produced six seeds. Distinctness within each ranking avoided filling the shortlist with several nearly identical threshold settings for the same K/filter pair.

Each seed retained its K, cloth setting and stage-one type thresholds. Colour thresholds were searched in three consecutive groups:

1. Warm colours: red's upper hue boundary $\{4, 6, 8\}$; orange's upper boundary $\{12, 15, 18\}$; yellow's upper boundary $\{30, 35, 40\}$; red's wrap-around lower boundary $\{168, 170, 172\}$; and the red/maroon brightness split $\{130, 150, 170\}$. The nominal Cartesian product contained 243 combinations.
2. Green, blue and purple: green's lower boundary $\{40, 45, 50\}$; green's upper boundary $\{85, 90, 95\}$; blue's lower boundary $\{95, 100, 105\}$; the blue/purple boundary $\{125, 130, 135\}$; and purple's upper boundary $\{160, 165, 168\}$. This also gave 243 nominal combinations.
3. Neutral colours and body selection: black's saturation split $\{40, 60, 80\}$; black's maximum brightness $\{70, 90, 110\}$; and minimum eligible body-cluster area $\{1, 2, 5\}\%$. This gave 27 nominal combinations.

The intervals retained the original logic: red occupies both ends of the hue scale; orange and yellow follow its low-hue interval; green and blue have separate lower bounds; blue and purple share a boundary. Hue

intervals must remain ordered, with strictly positive gaps between yellow and green, green and blue, and purple and high-hue red. Combinations violating these constraints were rejected. Existing unknown-colour gaps were therefore preserved rather than forcing every hue into a ball colour.

For each group, the current configuration was retained as a candidate. The group's best configuration under the same full-number ranking became the starting point for the next group. Two complete passes revisited warm, cool and neutral thresholds after later groups had changed. This was a sequence of exhaustive searches within small groups, not an exhaustive Cartesian search over every colour parameter jointly. It can miss a useful combination requiring simultaneous changes across groups, and it does not prove global optimality. Stage-one type thresholds were not revisited after colour tuning. Their remaining interactions with colour thresholds are another limitation of the staged search.

The actual second stage evaluated 5,526 valid combinations. Together with stage one this gave 7,326 scored trials. Trials repeated across groups, seeds or passes were retained in the CSV audit trail; this count is not the number of unique parameter vectors. Each trial recorded its parent configuration and stage so the selection sequence could be reconstructed. The minimum eligible cluster percentage was included for completeness, but with at most six clusters the largest non-light cluster often already exceeds all tested minimum areas. A parameter with no effect on these images should not be interpreted as precisely estimated.

3.4.6 Selected configuration and measured improvement

Table 8: Baseline and selected recognition parameters. Other colour parameters retain their original values.

Parameter	Baseline	Selected
K	2	6
Cloth filter (0=off)	0	0
Light minimum V	170	170
Light maximum S	165	125
Cue minimum light (%)	85	85
Solid minimum non-light (%)	70	80
Blue/purple hue boundary	130	125

Table 9: Exploratory recognition scores. Tuning uses Table1–6; the final column is full-number recognition on AllBalls, reported separately.

Configuration	Type	Type acc.	Number	Number acc.	Oracle	Ref. no.
Original baseline	76/86	88.37%	67/86	77.91%	70/86	16/16
Baseline + cloth filter	77/86	89.53%	69/86	80.23%	72/86	15/16
Best K/type grid	81/86	94.19%	73/86	84.88%	74/86	15/16
Best after colour search	81/86	94.19%	74/86	86.05%	75/86	16/16

The highest-ranked configuration used K=6, no cloth filtering, $V \geq 170$ and $S \leq 125$ for light clusters, an 85% light threshold for the cue ball, and an 80% non-light threshold for solids. Relative to the original rules, its blue/purple hue boundary moved from 130 to 125. Other colour thresholds remained unchanged. It was configuration 2153 in the saved run. The same configuration won both the full-number and type-first rankings after tie-breaking, so these names do not refer to two distinct selected models.

On the 86 matched tuning balls, type correctness increased from 76 to 81: 88.37% to 94.19%, a gain of 5.81 percentage points. Full-number correctness increased from 67 to 74: 77.91% to 86.05%, a gain of 8.14 percentage points. Five type errors and seven number errors were corrected relative to the baseline, with no previously correct predictions becoming incorrect in either category on these matched candidates.

The best stage-one configuration already produced 73 correct numbers. Colour tuning added one further correct number on Table1–Table6 and restored the stage-one reference score on AllBalls from 15/16 to 16/16. The latter restoration was observed after selection, not used to choose the hue boundary. Oracle-number correctness increased from 70/86 for the baseline to 75/86 for the selected configuration.

It remained only one ball above full-number correctness, indicating that supplying the correct type alone resolved little of the remaining number-recognition error.

The cloth filter improved the original thresholds slightly, but was not part of the selected overall configuration. Its benefit was therefore conditional on K and the thresholds. These results support keeping cloth filtering optional rather than automatically adding it to the main pipeline.

Table 10: Per-image counts before and after the recognition search. All comparisons use the same matched candidates.

Image	GT	Matched	Base type	Best type	Base no.	Best no.
Table1.jpeg	16	14	8	11	6	9
Table2.jpeg	16	16	15	16	13	15
Table3.jpeg	16	15	14	14	14	14
Table4.jpeg	15	14	14	14	13	13
Table5.jpeg	14	14	14	14	11	12
Table6.jpeg	13	13	11	12	10	11

Improvements were unevenly distributed. On Table1, correct type predictions increased from 8/14 to 11/14 and correct numbers from 6/14 to 9/14. On Table2, correct numbers increased from 13/16 to 15/16. Tables5 and 6 each gained one correct number, while Tables3 and 4 retained their baseline number counts. This per-image view prevents the pooled gain from being mistaken for an equally large improvement on every scene.

Using all 90 annotations as the denominator, correct detected-and-numbered balls increased from 67/90 (74.44%) to 74/90 (82.22%). Detection itself was unchanged; the four missed balls could not be recovered by recognition tuning. The 16/16 reference result also matched, rather than exceeded, the original baseline on AllBalls.

Table 11: Selected configuration's type confusion matrix on Table1–6. Rows are ground truth and columns are predictions.

Actual / predicted	White	Solid	Striped	Unknown
White	6	0	0	0
Solid	0	42	1	0
Striped	3	1	33	0

The selected configuration still made five type errors and twelve full-number errors among matched tuning detections. For example, some striped balls were predicted as white because their visible light region dominated the crop. Some purple balls were assigned another colour or returned unknown; two black balls were assigned yellow. These examples show that tuning global thresholds does not resolve all effects of ball orientation, shadows, highlights and the selection of a single representative body cluster. They also explain why most remaining number errors persisted when the true type was supplied.



Figure 3: One matched candidate from Table1, compared under the original baseline, the cloth-filter baseline, the best stage-one configuration and the final selected configuration. Only final masked ball crops are shown. Each panel reports K, cloth setting, predicted type and number. Green text denotes a correct prediction and red denotes an incorrect prediction against the matched annotation. Identical winning configurations are displayed once.

3.4.7 Reproducibility, outputs and remaining limitations

The search is run from the repository root with `./build-local/BallExperiments--recognition-grid` after building the `BallExperiments` CMake target. Search ranges and original defaults are defined in `experiments/recognitionSearchConfig.h`. An optional output directory argument allows separate runs to be preserved.

Outputs are written under `output/experiments/recognition_grid_search/`. The type and colour leaderboards contain every evaluated configuration, its complete parameter vector, ranking metrics, stage and parent identifier. The K-comparison table records both the original thresholds and the best stage-one thresholds for each K/filter pair. Other CSV files contain the seed shortlist, finalist configurations, pooled and per-image metrics, candidate-level predictions, type and number confusion matrices, detection counts, mask sizes and cached cluster features. Unmatched candidates have blank correctness fields rather than being silently counted as correct or incorrect classifications. A separate image is saved for each candidate, with the baseline and distinct finalist configurations placed side by side.

Report tables and the diagnostic figure are local snapshots produced by `python3experiments/writeRecognitionReport.py`; the report can compile without the experiment outputs. Rerunning the search does not automatically update those snapshots. The surrounding interpretation must also be reviewed if parameters or measured results change.

The main limitation is selection on a small, correlated image set. Thousands of threshold trials can fit properties specific to these balls, their orientations and the shared illumination. The measured gains should therefore motivate evaluation on fresh photographs before replacing the baseline. Neither the pooled tuning result nor the AllBalls reference check establishes performance on independent data. The search was also bounded to five K values, small threshold grids and two grouped colour passes; it establishes the best observed configuration under that procedure, not the best possible recognition system. No new production settings were applied as part of this experiment.

3.5 Tuning the Connected-Component Recognition Approach

The final recognition experiment investigated whether spatial measurements could improve the tuned colour classifier. The preceding search selected configuration 2153, which correctly classified 81/86 matched ball types and 74/86 numbers on Table1–Table6. Rather than replacing this classifier with an entirely different method, CCA was evaluated as an optional type-correction stage. Its purpose was to distinguish a substantial white stripe from smaller white markings and highlights using connectivity and boundary contact.

The earlier default CCA method reduced accuracy, but that result only tested one set of thresholds. This experiment separately investigated feature extraction, decision thresholds and which correction rules should be enabled. All changes remained in the independent experiment executable. The production CCA implementation and main application were not modified.

3.5.1 Fixed recognition base and control configurations

The primary search froze configuration 2153: K=6; light-cluster thresholds $V \geq 170$, $S \leq 125$; an 85% cue-ball light threshold; an 80% solid-ball non-light threshold; and a blue/purple hue boundary of 125. The other colour thresholds retained their original values. Extraction used the original Hough radius with zero padding and no cloth filtering. Neither the colour clusters nor body-cluster selection nor number thresholds were tuned again. CCA could change the type passed to number recognition, but could not change the cluster colours themselves.

Four controls were measured: the original K=2 classifier without CCA; that classifier with production CCA; the tuned K=6 classifier without CCA; and the tuned K=6 classifier with the original CCA thresholds. For the last control, the definition of a light cluster was aligned with the K=6 base ($S \leq 125$). It was not left at the original K=2 saturation threshold of 165. This alignment avoids constructing CCA features from a different light/non-light partition than the classifier whose decisions they modify.

Detection and one-to-one annotation matching were performed once per image and reused for all CCA settings. The run again contained 86 matched annotations out of 90 on Table1–Table6, plus 16 matched annotations on AllBalls. Unmatched detections were displayed but excluded from classification accuracy. Table1–Table6 provided tuning scores; AllBalls remained a separate reference check and did not influence ranking. The same limitations of correlated scenes and reuse of evaluation labels described in Section 3.4 apply here.

3.5.2 Constructing and measuring connected regions

K-means clusters contain pixels with similar BGR colours, regardless of their positions. CCA introduces spatial information after clustering. All cluster masks classified as light were combined with a binary OR operation, then restricted to the ball mask. The non-light mask was the complement of the light mask within the ball. Adjacent light pixels from different K-means clusters could therefore form one connected light region. Conversely, disconnected patches from one colour cluster could form several regions.

Eight-connected component labelling was applied separately to the light and non-light masks. A component's area was divided by the retained ball-mask area, rather than by the area of its own colour category. The features were the number of eligible light components, the largest eligible light area, the number of eligible non-light components and the largest eligible non-light area. Eligibility required a minimum component area of $\{0.5, 1, 2, 3\}\%$ of the ball.

The circular mask was eroded with an elliptical kernel, and the eroded mask was subtracted from the original mask to define an inner boundary band. Kernel sizes 3×3, 5×5 and 7×7 compared successively wider bands. The original erosion-border semantics were retained. Light boundary coverage was the percentage of band pixels belonging to the raw light mask:

$$C_\partial = 100 \frac{|M_{\text{light}} \cap B|}{|B|}, \qquad B = M_{\text{ball}} \setminus \text{erode}(M_{\text{ball}}).$$

An empty band returned zero coverage.

Small-component filtering affected component counts and largest-component statistics only. It did not erase pixels from the light/non-light masks, change their raw percentages, or change boundary coverage. This distinction preserved the original feature definitions and prevented an area-cutoff change from silently altering several measurements at once. No morphological opening or closing was added to the colour masks.

The four area cutoffs and three kernels gave 12 feature configurations. These were cached for each candidate after K-means, so rule-threshold trials reused the same spatial measurements. For the original K=2 control, the experimental defaults were compared with production CCA on every candidate: light and non-light masks, boundary mask, component counts, largest areas, boundary coverage and final hybrid

type all had to agree. This guarded against improvements caused by accidentally changing the baseline algorithm.

3.5.3 Decision rules and search space

The hybrid retained the original decision order: test a cue-ball correction, then a striped-ball correction, then a solid-ball correction, and otherwise retain the colour-only type. Each rule could be enabled independently, giving all eight subsets of the three rules, including CCA disabled. This comparison distinguished improvements from a useful subset of rules from improvements requiring the full hybrid.

The following threshold groups were searched:

1. **Cue rule.** Minimum raw light percentage $\{80, 85, 90, 95\}$; minimum largest light component $\{70, 80, 90\}\%$ of the ball; and minimum light-boundary coverage $\{60, 75, 90\}\%$. The rule also required zero non-light components above the selected area cutoff. These conditions gave 36 combinations. The rule is an override when its conditions hold, not a veto on every other prediction of white: if no override fires, the base type remains.
2. **Stripe rule.** Minimum largest light component $\{5, 10, 15, 20, 25, 30\}\%$; minimum light-boundary coverage $\{5, 10, 20, 30, 40, 50\}\%$; and minimum total non-light proportion $\{2, 5, 10, 15, 20\}\%$. These gave 180 combinations. The non-light requirement prevents an almost entirely light ball from being automatically assigned a stripe merely because its light region is connected to the edge.
3. **Solid rule.** Minimum total non-light proportion $\{60, 70, 80, 90\}\%$; maximum light-boundary coverage $\{5, 12, 20, 30\}\%$; maximum eligible non-light component count $\{1, 2\}$, with at least one required; and minimum largest non-light component $\{0, 50, 70, 85\}\%$. These gave 128 combinations. A zero largest-component threshold disabled this additional dominance constraint. Allowing two components and adding a dominance threshold were experimental extensions; the original rule required exactly one component and had no explicit largest-area condition.

For every feature setting and enabled-rule subset, the original rule thresholds supplied the starting point. Two passes searched the cue, stripe and solid groups in that order, skipping disabled groups. Each group used its full Cartesian product while holding the other groups fixed, and the current configuration remained a candidate. This was a grouped search, not a full joint Cartesian product over all rule thresholds. It may miss interactions requiring simultaneous changes in several groups, and two passes do not guarantee convergence to a global optimum.

For each rule, four of the eight subsets enable it. Consequently the run scored

$$4 + 12\,[8 + 2 \times 4(36 + 180 + 128)] = 33{,}124$$

trials: four controls, 96 initial seeds, and 33,024 grouped threshold trials. Repeated parameter settings remained in the audit trail. This total is therefore not a count of distinct models. No thresholds were selected separately for individual images or ball numbers.

Ranking first maximised correct full-number predictions, then correct type predictions. Remaining ties preferred fewer number regressions, fewer type regressions, fewer type changes, fewer enabled rules, less departure from the original CCA parameters, and finally the earlier trial ID. All regressions and changes were measured against configuration 2153. A type-first ranking reversed the first two criteria. The departure metric was only a deterministic tie-break: it summed absolute feature and threshold changes using the fixed normalisations documented in `experiments/ccaGridSearch.cpp`. It was not an additional measured recognition score or a statistical regularisation estimate.

3.5.4 Results and rule ablations

Table 12: CCA controls and selected configuration. Tuning scores use the 86 matched balls in Table1–6; the reference column reports numbers on AllBalls. The K=6 default-CCA control retains original CCA rule thresholds but aligns its light definition with configuration 2153.

Configuration	Type	Type acc.	Number	Number acc.	Ref. no.
Original K=2	76/86	88.37%	67/86	77.91%	16/16
Original K=2 + CCA	73/86	84.88%	64/86	74.42%	16/16
Tuned K=6	81/86	94.19%	74/86	86.05%	16/16
Tuned K=6 + default CCA	79/86	91.86%	72/86	83.72%	16/16
Tuned K=6 + selected CCA	82/86	95.35%	75/86	87.21%	16/16

Table 13: Best observed configuration for each enabled-rule subset on the K=6 base. Fixes and regressions compare with configuration 2153. Each subset is tuned separately.

Rules enabled	Type	Number	Type fixes/losses	No. fixes/losses	ID
None	81	74	0/0	0/0	3
Cue	81	74	0/0	0/0	19326
Stripe	81	74	0/0	0/0	19503
Cue + stripe	81	74	0/0	0/0	19900
Solid	82	75	1/0	1/0	20242
Cue + solid	82	75	1/0	1/0	20535
Stripe + solid	82	75	1/0	1/0	21008
Cue + stripe + solid	82	75	1/0	1/0	21661

With original K=2 recognition, default CCA reduced correct types from 76 to 73 and correct numbers from 67 to 64. With the K=6 light definition aligned to configuration 2153, the default rules still reduced type correctness from 81 to 79 and number correctness from 74 to 72. Thus simply attaching the existing hybrid to the improved colour classifier was not beneficial.

After tuning, the best observed configuration correctly classified 82/86 types (95.35%) and 75/86 numbers (87.21%). Both number-first and type-first selection chose configuration 20242. Relative to the tuned K=6 base, it corrected one type and its dependent number, with no regressions on matched Table1–Table6 candidates. Each accuracy increased by 1.16 percentage points. End-to-end number recall increased from 74/90 (82.22%) to 75/90 (83.33%), with detection unchanged. AllBalls remained at 16/16 correct types and numbers.

The cue-only and stripe-only ablations did not improve on the K=6 base. The solid-only rule achieved the best scores. Some larger tuned rule subsets achieved the same scores, but did not provide additional correct predictions. The tie-break favouring fewer enabled rules selected the solid-only model. This is evidence for a limited corrective use of the spatial feature, not for enabling every CCA override.

Table 14: Original CCA settings and selected CCA settings on the fixed K=6 base. For enable flags, 1 means enabled and 0 means disabled.

Parameter	Default CCA	Selected CCA
Minimum component area (%)	2.000000	2.000000
Erosion kernel size	5	5
Cue rule enabled	1	0
Stripe rule enabled	1	0
Solid rule enabled	1	1
Solid minimum non-light (%)	70	70
Solid maximum light boundary (%)	12	20
Maximum non-light components	1	1
Minimum largest non-light region (%)	0	0

The winning feature settings retained the original 2% component cutoff and 5×5 erosion kernel. The cue and stripe overrides were disabled. The solid correction required at least 70% non-light pixels, at most 20% light-boundary coverage, and exactly one non-light component above the area cutoff. No additional largest-non-light dominance threshold was needed. The decisive threshold change relative to the original

solid rule was raising its allowable light-boundary coverage from 12% to 20%. These correction thresholds are separate from the base classifier's 80% non-light threshold.

3.5.5 Interpreting the corrected ball

Table 15: Matched candidates changed by the selected CCA configuration. Candidate IDs refer to detections rather than annotated ball numbers.

Image	Candidate	True no.	Base no.	CCA no.	Base type	CCA type
Table1.jpeg	10	5	13	5	Striped	Solid

Only candidate 10 in Table1 changed under the selected correction. Its annotated number was 5, a solid orange ball. The K=6 colour classifier measured 21.83% light and 78.17% non-light pixels. Since the non-light proportion fell below the base solid threshold of 80%, it predicted striped, causing the orange body colour to be assigned number 13.

CCA found three light components above the 2% area cutoff, with the largest occupying 12.74% of the ball. It found one eligible non-light component, occupying 76.11% of the ball. Light pixels covered 17.68% of the boundary band. This failed the old solid correction's maximum boundary coverage of 12%, but satisfied the tuned maximum of 20%. Together with the 70% non-light requirement and single-component condition, this changed the type to solid and the dependent number to 5.

The interpretation is that a substantial connected body region and relatively little light at the boundary provided useful evidence for a solid ball, even though its total light proportion was too high for the colour-only rule. This is a feature-based explanation of this candidate, not proof that every similar mask corresponds to a solid ball.

The diagnostic crop also contains substantial parts of neighbouring balls in the tightly arranged Table1 scene. Its connected non-light region cannot therefore be assumed to represent only the target ball's body. A correct classification does not establish an accurate segmentation. In this case, the tuned rule may partly compensate for contamination from imperfect localisation rather than isolate a property of an unobscured solid ball.



Figure 4: The corrected Table1 candidate under the original K=2 control, original CCA, tuned K=6, default CCA on K=6, and the selected solid-only CCA correction. Each panel shows the masked ball and its light/non-light partition. White and orange represent interior light and non-light pixels; green and red in the mask represent light and non-light boundary pixels. Prediction text uses green for correct and red for incorrect. The mask colours are feature visualisations, not ground-truth segmentation labels.

3.5.6 Ceiling for type-only corrections and limitations

The preceding colour experiment achieved 75/86 oracle-number predictions when supplied with true types. The CCA experiment independently recomputed the same diagnostic with its fixed K=6 features and number rules. For these rules, a correct predicted number implies the correct type: 0 corresponds to white, 1–8 to solid, and 9–15 to striped. Consequently, 75/86 is an upper bound on full-number correctness achievable by changing only the type of these cached candidates. The selected CCA correction reached that measured ceiling. Further type-only changes cannot improve full-number correctness on this fixed feature set; the remaining eleven number errors require changes to colour interpretation, body selection, segmentation or extraction. This is not an upper bound on a different recognition pipeline, and it does not imply that all type errors have been corrected: four type errors remain.

The observed gain is small and entirely due to one candidate in one image. Searching over 33,124 trials on the same small tuning set creates a substantial opportunity to select a threshold that fits this example. The absence of regressions on these images is useful for comparison, but does not establish that the rule will be harmless on new ball orientations or lighting. An independent image set would be needed before treating the one-ball gain as a general improvement. The report therefore records the result as a promising targeted correction rather than a definitive replacement for the colour-only classifier.

3.5.7 Reproducing and inspecting the CCA experiment

After building the independent experiment target, run from the repository root:

```
./build-local/BallExperiments--cca-grid
```

An optional directory argument preserves separate runs. Feature ranges, rule thresholds and the frozen recognition base are defined in `experiments/ccaSearchConfig.h`. Outputs are written to `output/experiments/cca_grid_search/`.

The leaderboard records every trial's full CCA configuration, stage, parent, scores and regression counts. The ablation CSV selects the best observed model for each enabled-rule subset, and the seed CSV records each grouped search's starting and final IDs. Finalist configurations, per-image metrics, candidate predictions, confusion matrices, detection counts and cached features support inspection of the results. The number-ceiling CSV records the true-type diagnostic per image and per split. One diagnostic image per candidate compares distinct control/finalist configurations, with no individual ball subdirectories. Unmatched candidates have blank correctness fields and unavailable correctness indicators.

The report tables and figure are refreshed with `python3experiments/writeCCAReport.py`. They are snapshots rather than live links to the output directory. The surrounding prose contains run-specific counts and should be reviewed after changing the grid. No search result was automatically applied to the main application.

3.6 Watershed Segmentation for Crowded-Ball Detection

The first segmentation-based detection experiment investigated whether marker-controlled watershed segmentation could recover balls missed by the circular Hough detector in crowded regions. Unlike the preceding experiments, its primary objective was improved localisation rather than a different type or colour threshold. The method was implemented as an independent fixed prototype in `experiments/watershedDetection.cpp`, without replacing the detector used by the main application. No parameter grid was run.

Watershed separates image regions by growing regions from labelled markers. An additional ball therefore requires an additional suitable marker; merely running watershed from the existing Hough centres cannot recover a ball whose interior has no marker. The experiment followed the general distance-transform/marker approach illustrated in OpenCV's distance-transform and watershed tutorial, adapted to the known table geometry and existing Hough detections.

3.6.1 Auditing the baseline misses

The current detector was rerun on every image, including table masking, Gaussian smoothing, Hough detection and the existing pocket filter. Its centres were transformed into table coordinates and matched one-to-one to annotations within 15 table pixels. Raw Hough candidates were evaluated separately to identify whether the pocket filter removed a previously matched ball. The audit also recorded whether a missed annotation lay inside one of the image-derived crowded-region windows.

Table 16: Baseline missed annotations. Source coordinates are obtained by inverse perspective transformation for evaluation and visualisation only; they never supply watershed markers.

Image	Ball	Source x	Source y	Nearest error	Raw match	In ROI
Table1.jpeg	6	1403.6	149.7	33.47	No	Yes
Table1.jpeg	7	1431.9	202.7	33.74	No	Yes
Table3.jpeg	10	1474.9	101.2	55.43	No	Yes
Table4.jpeg	10	1474.9	101.2	52.49	No	Yes

On Table1–Table6, the current run matched 86 of 90 annotations and produced two false detections. The four misses were balls 6 and 7 in Table1 and ball 10 in both Table3 and Table4. All four were already unmatched before pocket filtering and all lay inside crowded windows. Thus this run did not support pocket filtering as the immediate explanation for these misses. The annotations' distances to their nearest filtered detections exceeded the 15-pixel matching tolerance, rather than simply requiring a different assignment of two otherwise eligible matches.

These counts were measured by rerunning the current source and agree with the current baseline in Table 3. An older saved validation CSV recorded seven false detections; it is retained as a historical artifact rather than mixed with these new comparison results. Recognition parameters had already been tuned using Table1–Table6; downstream recognition scores therefore remain exploratory rather than independent test estimates.

Inverse perspective transformation was used to display missed annotations at their approximate original-image positions. These points were used only for evaluation and diagnostic overlays. They did not influence window construction, foreground masks, distance peaks, markers, basin validation or proposal acceptance. The segmentation function accepts an image and baseline circles, not annotations.

3.6.2 Local windows and foreground construction

Pairs of filtered Hough circles were considered nearby when their centre distance was at most 1.4 times the sum of their radii. Transitive groups with at least two detections formed the initial crowded groups. Their circle bounding boxes were expanded by twice the median group radius and clipped to the image. Overlapping windows were merged, and every baseline circle centred in a resulting window supplied a Hough marker. The expected radius for new candidates was the median Hough radius in that window. Ten windows were processed across the seven images.

This local design preserves isolated detections and limits processing of unrelated cloth or table features. Its limitation is equally explicit: a completely missed group with no suitable nearby Hough detections, or an isolated missed ball outside the selected windows, cannot be recovered. The audit showed that this coverage limitation did not exclude any of the four missed annotations in this particular run.

Cloth colour was estimated separately in each window. Eligible samples lay inside the known table polygon, outside disks of 1.25 times each local Hough radius, and satisfied $35 \leq H \leq 95$, $S \geq 35$ and $V \geq 30$ in OpenCV HSV units. Samples were taken every two pixels in each direction. At least 40 eligible samples were required; otherwise the window was skipped without changing the Hough detections.

The samples were converted to OpenCV's encoded 8-bit Lab representation. The cloth representative was the component-wise median Lab vector. Let d_j be a sample's Euclidean distance from that representative. A pixel was provisionally foreground when its Lab distance exceeded

$$T = \mathrm{clip}_{[16,45]}\left(\mathrm{median}(d_j) + 2.5\,\mathrm{median}\,|d_j - \mathrm{median}(d_j)|\right).$$

These numerical distances apply to the encoded channel values and should not be interpreted as calibrated

perceptual colour differences.

Small foreground cores of radius $0.65r$ were retained around existing Hough centres. This reduced the chance that a known green ball disappeared entirely into a green-cloth mask. It did not solve the corresponding problem for a missed green ball with no Hough marker. A 3×3 elliptical closing operation filled narrow gaps, enclosed holes were filled using flood filling from a padded exterior, and the result was intersected with the table polygon. In a tightly packed group, this foreground can remain one merged object rather than one object per ball.

3.6.3 Markers, watershed and candidate acceptance

Three variants were compared:

1. **Hough baseline:** the unchanged filtered Hough circles.
2. **Hough-seeded watershed:** one marker per existing Hough detection, with no new detection proposals. This is a segmentation control: circles are not relocated and no Hough circle is removed.
3. **Hybrid watershed:** Hough markers plus additional distance-transform peaks that pass placement and basin checks.

The foreground's Euclidean distance transform measured distance to the nearest zero pixel. Local maxima were detected using rectangular dilation with a neighbourhood approximately 0.9 expected radii wide. Peaks had to reach at least 0.45 expected radii in distance. Connected peak plateaus were represented by an actual plateau pixel nearest their centroid, rather than assuming that a centroid necessarily lies on a peak.

Peaks were considered in decreasing distance order. A proposed marker had to be at least 1.15 times the larger radius from every existing Hough marker and previously accepted additional marker. It also had to lie at least 0.6 expected radii from the crop edge. These checks reduce duplicate markers and proposals truncated by local windows, but can reject a distance peak displaced towards a neighbouring ball.

Markers used distinct positive integer labels in a 32-bit image. Hough and additional foreground markers were disks of radius three pixels. Sure background was outside a 7×7 elliptical dilation of the foreground or outside the table polygon; the intervening collar remained unknown. Watershed was applied to the original BGR crop after 3×3 Gaussian smoothing with $\sigma = 0.8$. It was not applied directly to the floating-point distance transform. Separate seeded and hybrid label maps were saved for comparison.

For each additional marker, its labelled basin was intersected with the foreground mask. Acceptance required:

- Basin pixel area between 0.25 and 1.8 times the expected disk area.
- Circularity $4\pi A/P^2 \geq 0.35$, measured from the largest external contour's area A and perimeter P.
- Largest-contour bounding-box aspect ratio at most 2.2.
- At least 35% foreground support inside an expected-radius disk.
- No duplicate of an existing or newly appended circle, and passage through the same pocket filter used for baseline detections.

An accepted circle would use the distance peak as its centre and the local median Hough radius as its radius. A visible watershed basin's centroid was not substituted for an existing Hough centre, since partial occlusion can shift a visible-region centroid away from the physical ball centre. Existing circles and radii were preserved. Consequently, this prototype could add missed candidates but could not correct an existing inaccurate centre or delete a baseline false positive.

3.6.4 Detection and recognition results

Table 17: Current-run detection comparison on Table1–6. The 90 annotations and 15-table-pixel matching tolerance are identical for all methods. Error is mean matched centre distance in table pixels.

Method	TP	FP	FN	Precision	Recall	F1	Error
Hough baseline	86	2	4	97.73%	95.56%	96.63%	4.91
Hough-seeded watershed	86	2	4	97.73%	95.56%	96.63%	4.91
Hybrid watershed	86	2	4	97.73%	95.56%	96.63%	4.91

Table 18: Distance-peak outcomes across all seven images. Only peaks passing marker placement proceed to watershed-basin validation.

Outcome	Count
basin_area	1
basin_circularity	3
hough_duplicate	40
near_crop_edge	8

The prototype did not recover any real-image detections. Across all seven images it produced 52 eligible local distance maxima. Forty were rejected as duplicates of Hough markers and eight lay too close to a window edge. Four additional markers proceeded to basin validation: three were rejected for insufficient circularity and one for insufficient area. None was accepted. The resulting hybrid circle lists therefore equalled the baseline circle lists.

On Table1–Table6, all methods consequently had 86 true positives, two false positives and four false negatives. Precision was 97.73%, recall 95.56% and F1 96.63%. Mean matched-centre error was 4.91 table pixels. No annotated ball was recovered or lost. The Hough-seeded control was expected to retain baseline detection scores by design; the hybrid's unchanged score was the empirical negative result. AllBalls retained 16 matched detections with no false detections.

Table 19: Per-image baseline and hybrid counts. Recognition is fixed to the tuned K=6 classifier plus selected solid-only CCA. AllBalls is the reference image.

Image	Base TP	FP	FN	Hybrid TP	FP	FN	Correct no.
AllBalls.jpeg	16	0	0	16	0	0	16
Table1.jpeg	14	1	2	14	1	2	10
Table2.jpeg	16	1	0	16	1	0	15
Table3.jpeg	15	0	1	15	0	1	14
Table4.jpeg	14	0	1	14	0	1	13
Table5.jpeg	14	0	0	14	0	0	12
Table6.jpeg	13	0	0	13	0	0	11

Recognition was held fixed to configuration 2153 plus the selected solid-only CCA correction from Section 3.5. All methods used the standard circular extraction mask, not the watershed basin. This isolates detection proposals from an additional change to recognition segmentation. With no accepted additional circles, correct type and number counts remained 82/86 and 75/86 on matched Table1–Table6 balls, and 16/16 on AllBalls. End-to-end number recall remained 75/90 (83.33%). This experiment did not test whether substituting watershed masks for circular recognition masks would improve colour estimates.

3.6.5 Why the additional markers failed

The diagnostics showed a limitation of distance-transform seeding for these particular masks. In Table1, closely packed balls, shadows and foreground cleanup formed a broad connected foreground mass. Its distance transform had large interior peaks corresponding to the group shape, rather than a separate peak at every visible ball. The two extracted peaks were both close to existing Hough detections. In particular, there was no independent accepted marker for the missed green or maroon ball. The green-cloth similarity and occluded boundaries are plausible contributors, but the experiment did not separately quantify their causal effects.

In the Table3 example, the missed striped blue ball touched the detected black ball. Their combined foreground produced a distance maximum between the visible objects, near the already detected black ball. Duplicate suppression rejected it as an additional marker. A geometric peak in the merged silhouette therefore did not reliably identify the missing ball's centre. All four missed annotations lay inside processed windows, so simply enlarging the processing coverage would not resolve this observed marker problem.

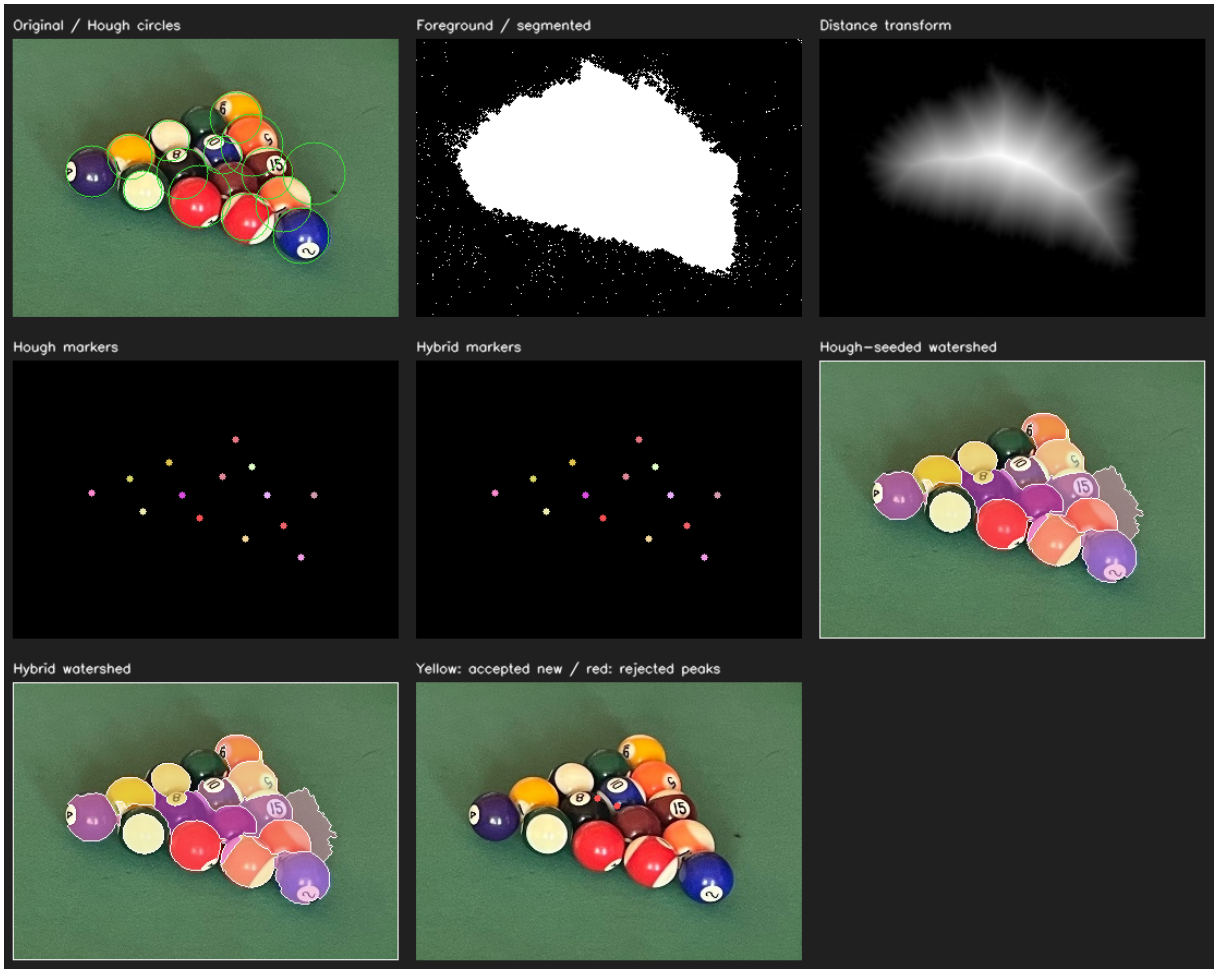


Figure 5: Table1 crowded-region diagnostics. The foreground mask merges the packed group, and the distance transform does not provide one independent marker per ball. Hough-seeded and hybrid watershed use the same surviving markers in this case. The final panel shows rejected distance peaks in red; no new circles were accepted. Watershed region colours are arbitrary labels and white lines are watershed boundaries.

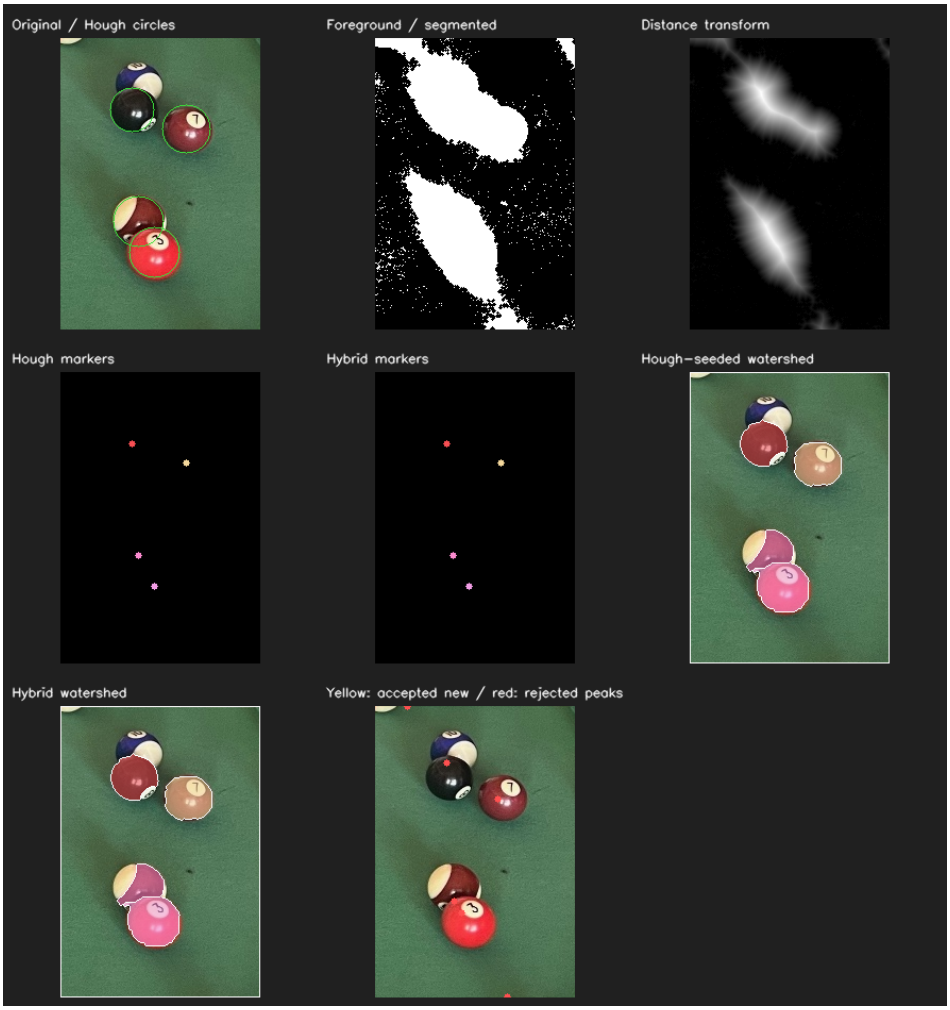


Figure 6: A Table3 window containing the missed striped blue ball next to a detected black ball. The merged foreground's distance peak does not provide a sufficiently independent new marker. The seeded label map consequently does not separate the missed ball as a new object.

The implementation was also checked on a synthetic scene containing three overlapping, differently coloured circles on a green background, with only two Hough circles supplied. The hybrid recovered one additional circle within three pixels of the known synthetic centre, preserved the two supplied circles exactly, and retained the expected radius. Additional checks covered empty or isolated detections, invalid image types and insufficient cloth samples. This confirms that the proposal path can function under controlled conditions; it does not establish reliability on the real crowded scenes.

3.6.6 Interpretation, outputs and reproducibility

The measured conclusion is that this fixed watershed prototype did not improve detection. It does not establish that every watershed-based method would fail. The limiting factors were the provisional foreground representation, the availability of independent markers, and the conservative acceptance policy. Specular highlights, stripes and neighbouring balls also create internal colour boundaries that need not coincide with object boundaries. Further work could investigate additional edge-based markers or fitting visible circular arcs, but those alternatives were not implemented in this comparison. Acceptance thresholds were not relaxed after inspecting the annotations in an attempt to force a positive result.

There are no pixel-level ground-truth masks in the provided data. Segmentation quality was therefore inspected visually; the experiment reports centre-detection and recognition scores rather than mask IoU or boundary accuracy. Watershed cannot directly reconstruct the hidden part of a ball, and a segmented visible region is not automatically an unbiased estimate of its complete circular boundary.

The experiment runs from the repository root with:

```
./build-local/BallExperiments--watershed
```

An optional output-directory argument preserves separate runs. The default directory is `output/experiments/watershed_detection/`. The source options are in `experiments/watershedDetection.h`. Saved files include pooled/per-image metrics, the raw-versus-filtered ground-truth audit, local-window metadata, every peak's acceptance or rejection reason, matched recognition predictions, and recovered/lost annotation identities. Parameter values are exported to CSV and the prototype procedure is described in a companion Markdown file. Each image has one full detection comparison and one diagnostic sheet per crowded window, rather than individual ball subdirectories.

Report snapshots are refreshed using `python3experiments/writeWatershedReport.py`. The surrounding discussion contains run-specific observations and must be reviewed if the implementation or results change. The main detection and recognition pipelines remain unchanged.

3.7 Segmentation-First Routing to Two Hough Searches

The preceding watershed experiment depended on existing Hough detections to locate crowded windows and on distance-transform maxima to propose missing markers. Its failure to recover the striped blue ball above the black ball in Table4 motivated a different order of operations. The next experiment constructed foreground components before running Hough, identified potential clusters from their area and shape, and used different Hough settings inside and outside those windows. A separate watershed check then assessed whether the newly generated markers improved candidate validation.

Three methods were compared: the current standard Hough detector; segmentation-guided Hough with circular-arc validation; and the same guided candidates followed by watershed-basin validation. All methods used the same known table polygon and the same final pocket filter. Detection proposals were generated before loading ground-truth annotations. No annotated ball centre was used to select a component, seed a marker, or accept a circle. The main application was not changed.

3.7.1 Foreground components before any Hough call

Foreground was estimated over the table using 128×128 image tiles. Within each tile, every second pixel in each direction was eligible as a cloth sample when its HSV values satisfied $35 \leq H \leq 95$, $S \geq 35$ and $V \geq 30$. At least 30 samples were required. The component-wise median of their encoded 8-bit Lab values

supplied the local cloth colour. The foreground-distance threshold was the median sample distance plus 2.5 times its median absolute deviation, clipped to $[16, 45]$, as in the preceding experiment. A global cloth estimate provided a fallback for tiles without enough samples. If no global model was available, no cluster was declared and standard Hough was retained everywhere.

No Hough-based exclusion disks or foreground cores were used at this stage. This distinguishes the experiment from the earlier Hough-first watershed prototype. Local cloth estimation can still confuse a green ball with the cloth, but an isolated ball omitted from the foreground is not automatically deleted: standard Hough still examines the original table-masked image.

The foreground construction inset the table polygon with a 41×41 elliptical erosion to suppress rail and shadow connections. A 3×3 elliptical closing filled small gaps, enclosed holes were filled, and the mask was restricted to the inset table. Components smaller than 80 pixels were discarded. An initial engineering run with an 11-pixel inset had connected foreground along rails into nearly table-wide search windows. Its outputs were retained separately; the reported configuration uses the 41-pixel inset. This adjustment was based on inspecting the mask failure, but the experiment remains exploratory and should not be presented as an untouched held-out evaluation.

An image-coordinate size guide was constructed without Hough detections: the expected radius increased linearly from 15 to 30 pixels between the minimum and maximum source-image y coordinates of the table corners. This used the existing radius bounds and camera geometry, not annotated ball sizes. It is a heuristic; actual apparent radii need not follow that exact linear relation.

A foreground component was considered a potential cluster when its area was at least 1.6 times the expected disk area, or when its bounding-box aspect ratio was at least 1.45 and its area was at least 0.8 expected disk areas. Components covering over 10% of the table, or long rail-like components with aspect ratio above four and more than 60% of pixels in the table's boundary band, were excluded. A window extended each selected component's bounding box by 1.25 expected radii. Overlapping windows were merged. The final run produced 32 windows across seven images; some represented shadows or single balls rather than actual multi-ball clusters.

3.7.2 Routing standard and permissive circle searches

Standard Hough retained the original settings: $dp = 1.5$, minimum centre distance 18 pixels, upper Canny threshold 75, accumulator threshold 35, and radius bounds 15–30 pixels. It was computed on the unmodified table-masked image and then routed by candidate centre. This avoids introducing artificial image edges by painting cluster interiors black before circle detection.

Standard candidates centred in the interior of a cluster bounding box were replaced by results from its local permissive search. An overlap margin of half an expected radius retained standard candidates near the region boundary, while the padded local crop covered the same area. Thus routing did not intentionally leave a gap between the two searches.

The local search retained $dp = 1.5$, Canny threshold 75 and the original 5×5, $\sigma = 1.5$ preprocessing, but reduced the accumulator threshold to 22 and minimum centre distance to 12 pixels. Its radius bounds were $\max(12, \lfloor 0.7 r_e \rfloor)$ and $\min(32, \lceil 1.6 r_e \rceil)$, where r_e is the window's size guide. This widens the plausible range modestly instead of treating occlusion as evidence that the physical ball radius has become much smaller. The exact bounds for each window were saved to CSV.

Local searches used the original colour crops, not binary foreground images. Foreground was a routing and support cue; Hough still measured image edges. Each proposal was checked using 72 samples around its predicted circumference. A sample was supported when a Canny edge was within 2.5 pixels and the absolute cosine between its Sobel gradient direction and the circle's radial direction was at least 0.6. This reduces support from edges crossing a proposed circle tangentially or from unrelated internal markings.

At least 30% of circumference samples had to be supported. A contiguous supported arc of at least 75 degrees was also required, allowing isolated one-sample gaps to be bridged for the arc-length calculation only. The raw support percentage was not increased by that bridging. At least 35% of the proposed disk had to overlap foreground, and the centre had to lie inside the table. These tests allow a partly occluded circle to lack edge evidence along its hidden side.

Candidates passing these checks were ordered by $0.7 \times$ edge support $+ 0.3 \times$ arc angle$/360$. A candidate

was considered a duplicate only when its centre was closer than 0.8 times the smaller radius to an already selected circle. This replaces the earlier distance-peak experiment's much wider 1.15-larger-radius exclusion, which could suppress occluded neighbours. Standard detections retained by routing were selected first. If a window accepted no local proposal, available standard circles in that window were retained as an explicit empty-search fallback. This decision did not consult ground truth.

3.7.3 Separate watershed ablation

The optional watershed stage used accepted circle centres as distinct markers. This directly addresses the earlier missing-marker limitation: the new markers came from local image-edge evidence, not from maxima of the merged foreground silhouette. Small cores of half a detected radius were added to the watershed foreground around these accepted centres, after cluster selection had already finished. Foreground markers had radius three pixels; background lay outside a 7×7 foreground dilation or outside the table. Watershed used the original BGR crop smoothed with a 3×3 Gaussian, $\sigma = 0.8$.

For each accepted local candidate, the resulting label area overlapping the watershed foreground had to fall between 0.15 and 2.0 times its expected disk area. This was a candidate-rejection check, not a new centre-estimation method. Standard detections and explicit fallback detections were not removed by this check. Centres and radii were not changed. Comparing the guided variant before and after this check isolates this particular contribution of watershed; it does not test every possible use of watershed contours or masks.

3.7.4 Table4 recovery and overall results

Table 20: Table4: recovering ball 10 does not imply an improved overall detection result; other lost matches and false candidates must also be counted.

Method	TP	FP	FN	Ball 10 recovered	Correct no.
Standard Hough	14	0	1	No	13
Segmentation-guided Hough	14	2	1	Yes	12
Guided Hough + watershed	14	2	1	Yes	12

The guided search recovered the previously missed ball 10 in Table4. Its detected source centre was $(1479.75, 104.25)$ with radius 19.5 pixels. After perspective transformation its centre error was 7.00 table pixels, within the fixed 15-pixel matching tolerance. The watershed check retained this candidate. Thus visible circular-edge evidence could recover the ball that the earlier distance-peak marker strategy missed. The fixed recognition pipeline correctly predicted striped for this candidate, but returned an unknown number; recovery of its location did not imply correct number recognition.

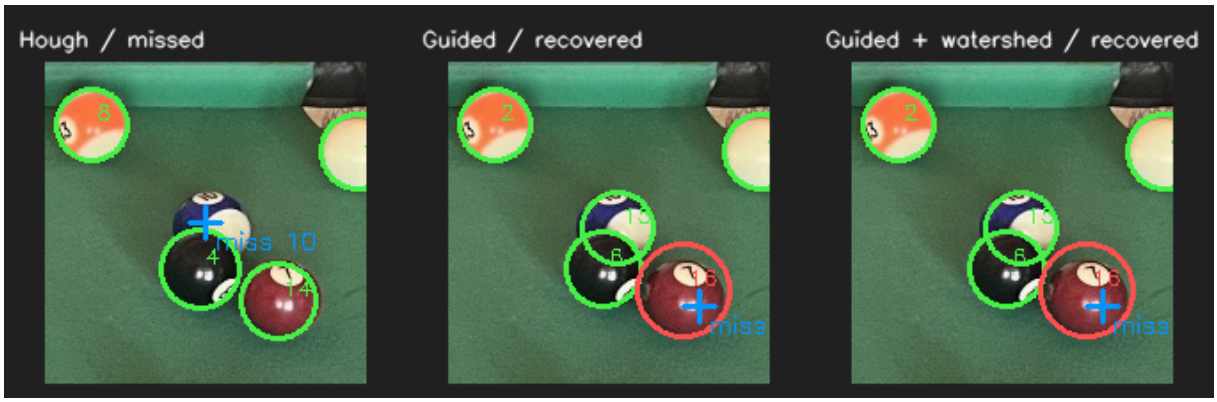


Figure 7: Evaluation-only close-up centred on the missed Table4 annotation. The standard detector misses ball 10, while the local permissive search recovers it and supplies a separate watershed marker. Green circles are matched detections, red circles are false detections and the blue cross marks a missed annotation. Ground truth selected this display crop after detection; it did not determine any proposal or marker.

However, the same Table4 run lost the baseline match to ball 7 and introduced additional false detections. Before watershed it produced 14 true and two false detections, compared with the baseline's 14 true and zero false detections. The watershed area check retained the same 16 circles in Table4. This is a change in which balls were found, not an overall Table4 detection improvement.

Table 21: Segmentation-first detection comparison on Table1–6. Recoveries and losses are annotated-ball identities relative to standard Hough; correct-number counts use the same fixed K=6/CCA recognition pipeline for all methods.

Method	TP	FP	FN	Precision	F1	Recovered/lost	Correct no.
Standard Hough	86	2	4	97.73%	96.63%	0/0	75
Segmentation-guided Hough	86	16	4	84.31%	89.58%	3/3	71
Guided Hough + watershed	83	11	7	88.30%	90.22%	3/6	69

Table 22: Annotation identities recovered or lost by the segmentation-guided Hough variant before the optional watershed check.

Image	Ball	Change
Table1.jpeg	7	recovered
Table1.jpeg	9	lost
Table3.jpeg	10	recovered
Table4.jpeg	7	lost
Table4.jpeg	10	recovered
Table6.jpeg	10	lost

Across Table1–Table6, guided Hough recovered three previously missed annotations: ball 7 in Table1 and ball 10 in Table3 and Table4. It also lost three previously matched annotations: ball 9 in Table1, ball 7 in Table4 and ball 10 in Table6. Total true positives therefore remained 86, with four misses. False detections increased from two to 16, reducing precision from 97.73% to 84.31% and F1 from 96.63% to 89.58%. The mean matched-centre error also increased from 4.91 to 5.29 table pixels.

Watershed reduced the guided variant's false detections from 16 to 11, but removed three additional correct matches. It therefore produced 83 true positives, seven misses, precision 88.30% and F1 90.22%. It was slightly better than unfiltered guided Hough in F1, but worse than standard Hough. On AllBalls, guided Hough retained 16 true detections while adding four false detections; watershed retained 14 true and three false detections, losing two reference balls.

Table 23: Per-image true and false detections. AllBalls is the reference image, reported separately from the six comparison images.

Image	Base TP	FP	Guided TP	FP	Guided+WS TP	FP
AllBalls.jpeg	16	0	16	4	14	3
Table1.jpeg	14	1	14	4	14	1
Table2.jpeg	16	1	16	5	14	4
Table3.jpeg	15	0	16	4	16	3
Table4.jpeg	14	0	14	2	14	2
Table5.jpeg	14	0	14	0	13	0
Table6.jpeg	13	0	12	1	12	1

Recognition was frozen to configuration 2153 plus the selected solid-only CCA correction, using ordinary circular masks. The baseline correctly classified 82 types and 75 numbers among its 86 matched balls. Guided Hough produced 78 correct types and 71 numbers, while guided Hough with watershed produced 75 types and 69 numbers. Changed local centres and radii can alter extraction and colour proportions even for an annotation still matched by both detectors. These scores therefore measure the downstream effect of the complete detection alternatives, not only the identities of recovered balls. Recognition thresholds were previously tuned on Table1–Table6, so these remain exploratory comparison scores rather than independent recognition test estimates.

3.7.5 Interpretation and reproducibility

The experiment supports the narrow hypothesis that a more permissive local edge search can supply a missing marker for the visible ball 10/ball 8 pair. It does not support replacing the current detector with this complete prototype. Segmentation sometimes groups shadows or single balls as clusters, local Hough accepts plausible circular fragments from background and internal features, and changing an existing crop can lose an otherwise acceptable annotation match. The watershed area check also rejects some correct markers because colour boundaries within a ball are not identical to object boundaries. These costs must be counted alongside the visually convincing recovery.

A stronger follow-up could preserve confirmed standard centres in a cluster and use local proposals primarily to supplement missing balls, with stricter background and arc-consistency tests. That would be a different comparison and was not silently substituted for the requested two-route experiment. No segmentation IoU is reported because the dataset supplies centres and numbers rather than pixel-level masks.

The command is:

```
./build-local/BallExperiments--segmentation-hough
```

An optional output path preserves separate runs. Options are defined in:

```
experiments/segmentationGuidedDetection.h
```

The default output directory is:

```
output/experiments/segmentation_guided_detection/
```

It contains pooled/per-image scores, per-annotation coverage and recovery/loss records, candidate predictions, full option values, cluster windows and routing metadata, radial edge/arc evidence, and watershed acceptance/rejection reasons. Diagnostic sheets show the foreground before Hough, routing interiors, raw local proposals, accepted local circles and watershed labels. Recovered annotations receive additional close-up comparisons for evaluation only.

Synthetic checks confirmed that changing the local Hough threshold did not change the foreground or routing mask, that accepted local candidates met their radial-arc criteria, and that watershed output was a subset of the guided circles. Empty local searches and absent cloth models were checked for baseline

fallback. All exported pooled/per-image scores and annotation changes were reconciled with predictions, and the standard control matched the preceding experiment's centres and recognition results.

Report tables and the Table4 figure are refreshed with:

```
python3experiments/writeSegmentationGuidedReport.py
```

The surrounding prose records this run's settings and results and needs review after a rerun with different options. All implementation changes remain in the experiment target; production detection and recognition functions and their parameters are unchanged.

3.8 Final Experimental Recommendation

The experiments support retaining the original masked, pocket-filtered Hough detector and recommending the tuned K=6 recognition configuration with the solid-only CCA correction. This is the strongest observed combination of localisation, type and number recognition among the evaluated alternatives. The recommendation is based on pooled results and recorded regressions, rather than selecting a method because it recovers one visually convincing example. Table1–Table6 were used to tune recognition, so the recommended recognition scores remain exploratory.

Table 24: Recognition progression with the original Hough detections fixed. All counts are over the same 86 matched balls in Table1–Table6.

Configuration	Correct type	Type accuracy	Correct number	Number accuracy
Original K=2 baseline	76/86	88.37%	67/86	77.91%
Tuned K=6	81/86	94.19%	74/86	86.05%
Tuned K=6 + solid-only CCA	82/86	95.35%	75/86	87.21%

The recommended settings are:

- **Detection:** retain $dp = 1.5$, minimum centre distance 18, upper Canny threshold 75, accumulator threshold 35, radii 15–30, the known table mask and the existing 25-table-pixel pocket exclusion.
- **Extraction:** retain the detected radius, with zero padding and no cloth filtering.
- **Clustering:** use six BGR colour clusters with the existing fixed-seed K-means implementation.
- **Light definition:** require $V \geq 170$ and $S \leq 125$, consistently in type classification, body-cluster selection and CCA.
- **Base type:** require at least 85% light for the cue ball or at least 80% non-light for a solid; otherwise retain the existing striped/unknown rules.
- **Number recognition:** use blue hue $100 \leq H < 125$ and purple $125 \leq H < 165$; retain the other original number thresholds and the minimum eligible cluster area of 2%.
- **CCA:** retain eight-connected components above a 2% ball-area cutoff and a 5×5 erosion boundary band. Enable only the solid override: at least 70% non-light pixels, at most 20% light-boundary coverage and exactly one eligible non-light component. Disable cue and stripe overrides; otherwise retain the base type.

Relative to the original K=2 baseline, this combination increased correct types by six and correct numbers by eight. Of these gains, the targeted CCA correction accounted for only one additional correct type and number beyond K=6. On AllBalls, the combination retained the original baseline's 16/16 correct types and numbers. Considering all 90 comparison annotations, including the four missed detections, correct types corresponded to 82/90 (91.11%) and correct numbers to 75/90 (83.33%). These are detected-and-correct recall measures, not precision measures: false detections must still be reported separately.

The other experiments informed the following exclusions. Padding admitted additional background and generally worsened number recognition. Cloth filtering slightly improved some original-threshold results,

but was not part of the best K=6 combination and could remove real green-ball pixels. The original full CCA rule set reduced accuracy; only the tuned solid correction was retained in the recommendation. The distance-peak watershed prototype recovered no detections. Segmentation-first Hough recovered three previously missed annotation identities, including ball 10 in Table3 and Table4, but lost three others and increased false detections from two to 16. Its watershed check reduced false detections to 11 but also reduced true positives from 86 to 83. These alternatives therefore did not justify replacing the standard detector.

There is a distinction between an experimental recommendation and application integration. The main application's current configuration still requests K=2 and calls the original type classifier, using the original light and solid thresholds. It does not invoke the selected solid-only CCA correction. The K=6 and CCA results above were implemented and measured in the independent experiment executable. No claim is made that the main application already produces the recommended recognition scores.

The next practical step is consolidation and evaluation on fresh images, rather than assuming that another processing stage will necessarily improve the result. This report records the selected experimental configuration, its limits, and the unsuccessful alternatives so that later integration can be checked against the same settings and evidence.

3.9 Qualitative Analysis and Failure Cases

Visual inspections revealed several recurring challenges.

Overlapping and touching balls. When balls are closely arranged, their visible boundaries may merge or be partially obscured. This can result in missed circles, imprecise centres or incorrect segmentation caused by pixels from neighbouring balls.

Ball orientation. A striped ball does not necessarily display equal amounts of white and colour. Depending on its orientation, the observed colour proportions may resemble either a solid ball or the white cue ball. This limitation is inherent to classification based on a single two-dimensional view.

Lighting and shadows. Highlights may cause coloured areas to appear white, while shadows can make coloured regions appear dark. Because classification depends on representative cluster colours and fixed HSV thresholds, these variations can change the predicted ball type.

Detection errors propagated into recognition. An inaccurate circle may include background pixels or exclude part of the ball. K-means then segments an imperfect crop, and the resulting percentages may no longer represent the actual ball.

These observations demonstrate why good localisation performance does not guarantee equally strong recognition performance.

4 Solution Analysis and Limitations

4.1 Limitations

Our method performs well on the supplied dataset, but several design choices limit its generalisation.

First, the fixed table mask and transformation depend on a known camera viewpoint. A different viewpoint would require new corner coordinates or an automatic table-localisation stage.

Second, a single set of Hough parameters may not be suitable for images containing balls with substantially different apparent sizes, lighting conditions or resolutions.

Third, the original two-cluster segmentation gives a coarse description of balls with a coloured body, white areas, printed markings, shadows and highlights. K=6 improved the observed recognition results when its thresholds were tuned jointly, but fixed colour rules and the choice of one body cluster still caused errors. More clusters alone did not improve the baseline.

Finally, the recognition and CCA parameters were selected using the six comparison images. Their best scores cannot be treated as independent test accuracy. The images share a camera and some unchanged

ball positions, and the one-ball CCA gain is too small to establish general reliability. New photographs would be needed to evaluate generalisation.

4.2 Possible Improvements

The experiments identify concrete limits without requiring another technique to be added to the current solution. Padding and cloth removal were not consistently beneficial. Segmentation-guided Hough recovered visible missed balls, but its false candidates and lost baseline matches outweighed those recoveries. Watershed labels alone were not evidence of improved detection.

The immediate recommendation is therefore to retain the standard detector and consolidate the strongest experimentally tested recognition settings. Further development should first be evaluated on new images, with attention to the remaining body-colour errors and crowded-ball crops. More elaborate boundary fitting or segmentation is future work, rather than a demonstrated improvement omitted from the reported solution. Consolidation into the main application has not been performed as part of this report update.

5 Conclusion

This project implemented a classical computer vision pipeline for pool-ball localisation and recognition using geometric information, circular Hough detection, perspective transformation and colour-based segmentation.

The standard detector was retained as the recommended localisation method. In the consistently rerun comparison it found 86 of 90 annotated balls, with two false detections, giving precision 97.73%, recall 95.56% and F1 96.63%. Segmentation-based alternatives could recover ball 10 in Table3 and Table4, but did not improve aggregate detection performance.

The strongest observed recognition combination used K=6 with tuned light, solid-percentage and blue/purple thresholds, followed by a solid-only CCA correction. It correctly classified 82/86 types (95.35%) and 75/86 numbers (87.21%), compared with the original baseline's 76/86 and 67/86. Correct detected-and-numbered balls represented 75/90 annotations (83.33%). These recognition scores are tuning results on the supplied comparison images, not independent estimates of unseen-image accuracy.

The final recommendation is the original detector with this tuned recognition configuration, without radius padding, cloth filtering, or segmentation-based detection. The main application still runs the original K=2 classifier; the recommended recognition configuration is available in the separate experiment executable and has not yet been integrated into that application.

The experiments demonstrate both useful improvements and the value of retaining a simpler method when additional processing causes regressions. They also show why an individually recovered ball, or an apparently good segmentation boundary, must be assessed alongside false detections, lost matches and downstream recognition errors.

6 Appendix

6.1 Use of Generative AI

6.2 Resources